

Applied Data Science Capstone

Predicting the Successful Landing of the Falcon 9 First Stage

SpaceX Launch Records · Machine Learning · Interactive Analytics

IBM Data Science Professional Certificate

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Executive Summary

● Methodologies

Data collection via SpaceX REST API and Wikipedia web scraping; data wrangling; EDA with SQL and visualization; interactive maps (Folium) and dashboard (Plotly Dash); predictive classification models.

● Key Results

Overall landing success rate of ~67%. Success improves with flight experience and varies strongly by orbit and launch site. Four classifiers reached ~83% test accuracy in predicting first-stage landing.

Goal: determine if the Falcon 9 first stage will land, enabling launch-cost estimation (a reused stage costs ~\$62M vs ~\$165M).

Introduction

Background

- SpaceX advertises Falcon 9 launches at \$62M; other providers cost up to \$165M.
- Much of the saving comes from reusing the first stage.
- If we can predict whether the first stage lands, we can estimate launch cost.
- This is valuable for any company competing to bid against SpaceX.

Questions to answer

- What factors determine a successful landing?
- How do payload, orbit and launch site affect success?
- How has success rate changed over the years?
- Which classification model best predicts landing?

Methodology

1

Data Collection

SpaceX REST API + Wikipedia
web scraping

2

Data Wrangling

Cleaning, one-hot encoding,
landing label

3

EDA

SQL queries + visual
exploration

4

Interactive Viz

Folium maps + Plotly Dash
dashboard

5

Predictive Analysis

LogReg, SVM, Decision Tree,
KNN + GridSearch

Data Collection — SpaceX API

- 1. GET request to the SpaceX REST API (/launches/past)
- 2. Decode JSON response into a DataFrame with `pd.json_normalize()`
- 3. Query helper endpoints (rocket, payload, launchpad, cores) to enrich rows
- 4. Filter to Falcon 9 launches; replace missing PayloadMass with the mean
- 5. Export the cleaned dataset to CSV

API → JSON
↓ `json_normalize`
DataFrame
↓ `enrich + clean`
90 Falcon 9 launches

Data Collection — Web Scraping

- 1. HTTP GET on the Wikipedia 'List of Falcon 9 and Falcon Heavy launches' page
- 2. Parse the HTML with BeautifulSoup
- 3. Locate the launch records table (wikitable)
- 4. Extract column headers, then iterate rows to collect each launch record
- 5. Build a DataFrame and export to CSV

```
Wiki page  
↓ requests.get  
HTML  
↓ BeautifulSoup  
↓ find table + rows  
DataFrame
```

Data Wrangling

- Computed launches per site, per orbit, and per landing outcome with `value_counts()`.
- Landing outcomes (True ASDS, True RTLS, True Ocean ...) were mapped to a binary label.
- Class = 1 for a successful landing, Class = 0 for an unsuccessful one.
- Resulting training label has 60 successes and 30 failures (mean success ≈ 0.667).

60

successful
landings (Class = 1)

30

failed
landings (Class = 0)

EDA with Data Visualization — Approach

- Scatter plots: Flight Number vs Launch Site / Orbit, and Payload vs Launch Site / Orbit, to reveal how experience and payload relate to success.
- Bar chart: success rate per orbit type, to compare orbits directly.
- Line chart: yearly average success rate, to expose the long-term trend.

EDA with SQL — Queries Performed

- Unique launch sites and launches beginning with 'CCA'.
- Total and average payload mass (e.g. by booster version F9 v1.1).
- First successful ground-pad landing date; boosters with success on drone ship in a payload range.
- Counts of mission outcomes; boosters that carried the maximum payload.
- 2015 failed landing records and a ranked count of outcomes between 2010-06 and 2017-03.

Interactive Visual Analytics

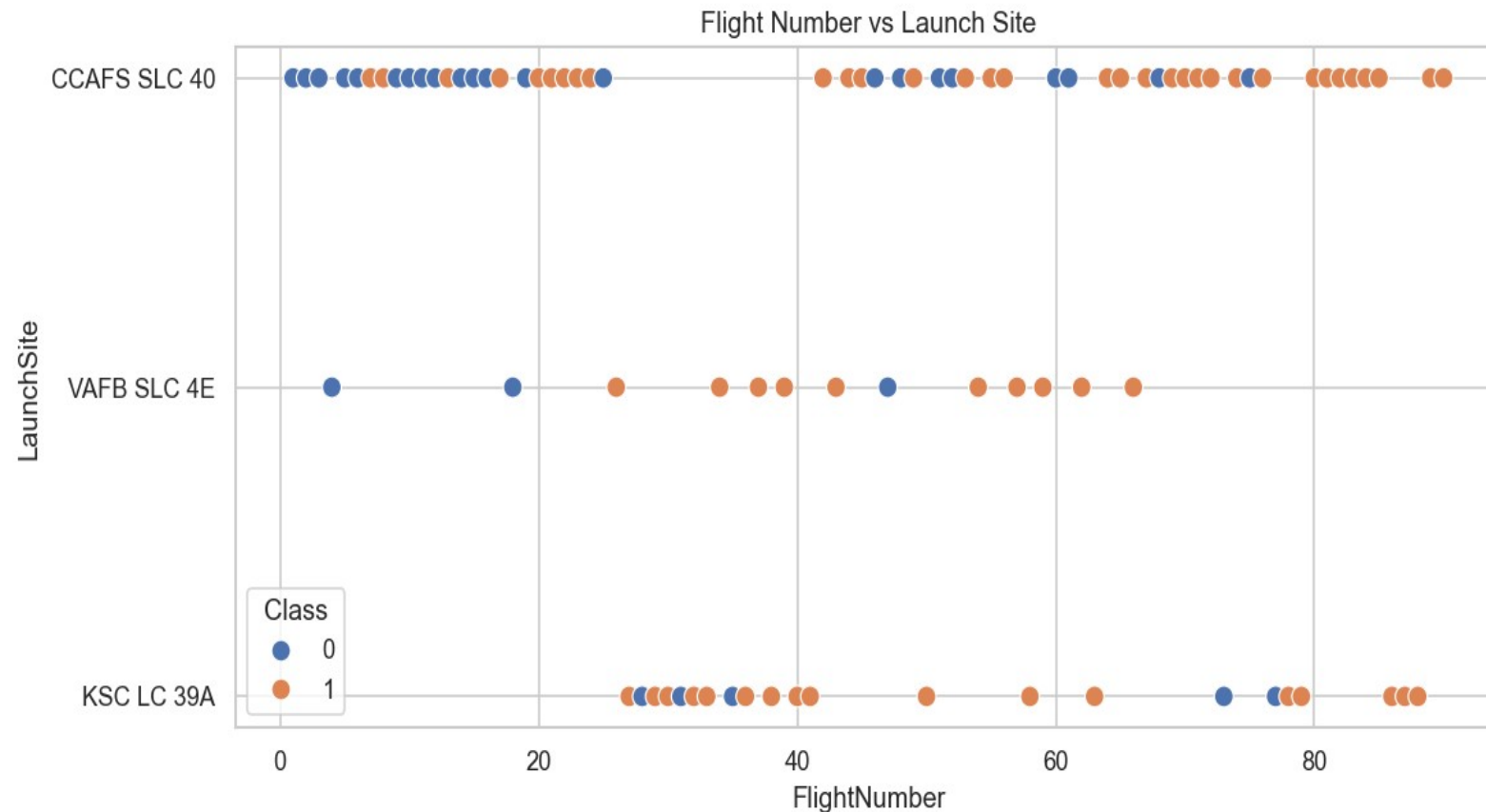
Folium Maps

- Marked all launch sites on an interactive map.
- Colored markers/clusters by success vs failure.
- Measured proximity to coast, railways and cities.

Plotly Dash Dashboard

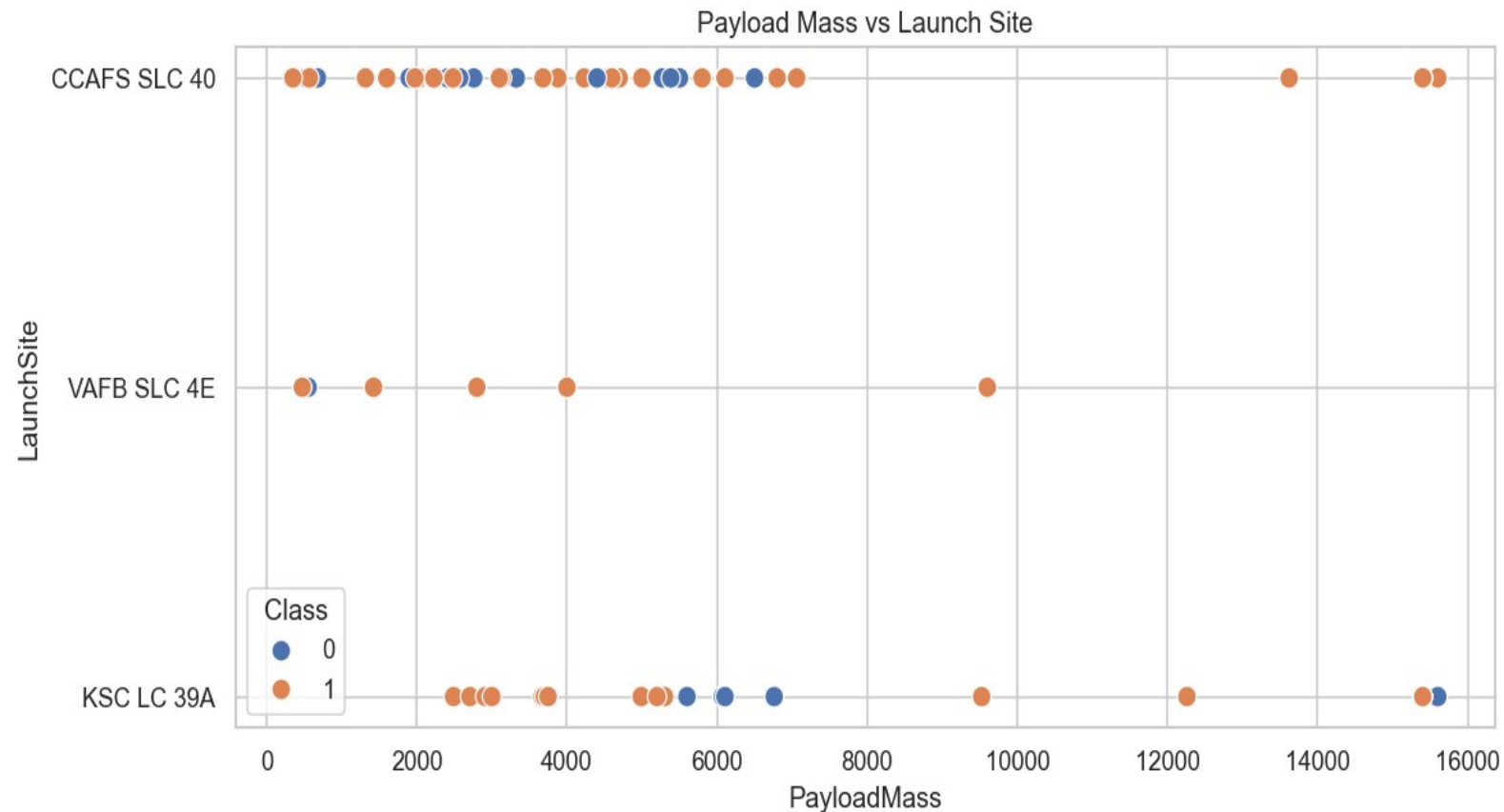
- Dropdown to select launch site.
- Pie chart of successful launches.
- Payload-range slider + payload-vs-outcome scatter.

EDA Results — Flight Number vs Launch Site



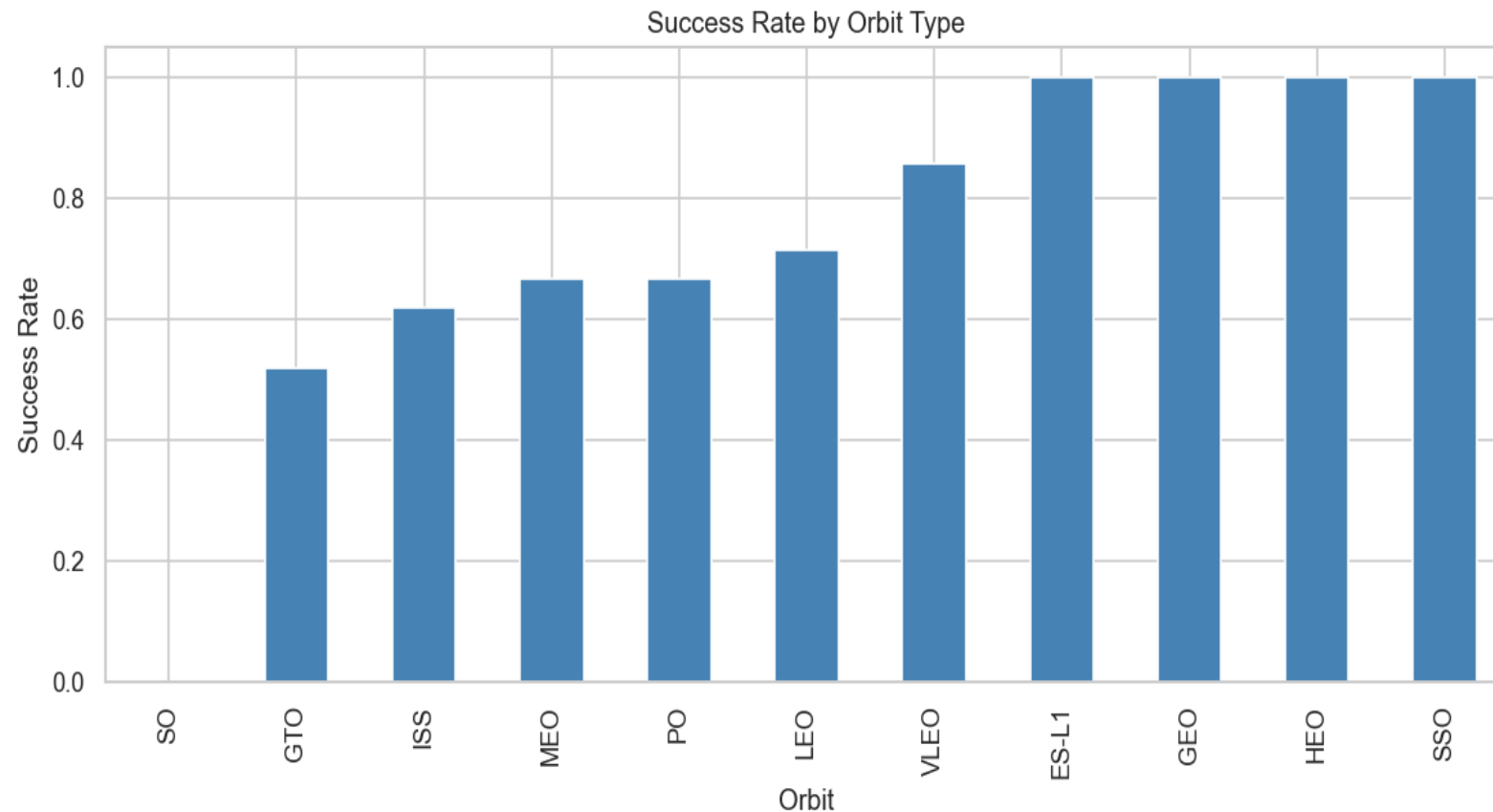
Later flights (higher experience) show more successes. CCAFS SLC 40 carries the most launches across all phases.

EDA Results — Payload vs Launch Site



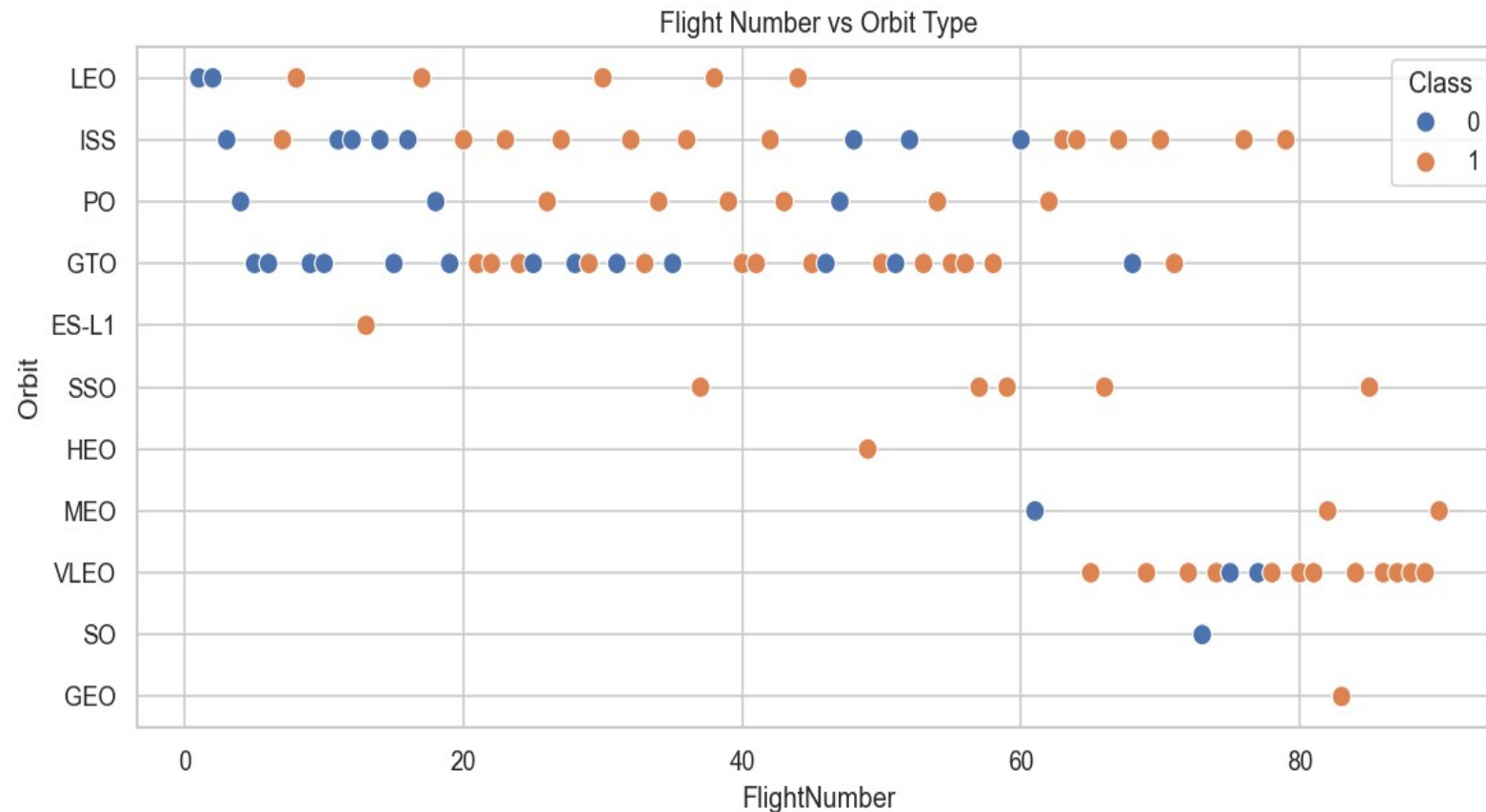
Heavier payloads are associated with more successful landings, especially at KSC LC 39A.

EDA Results — Success Rate by Orbit



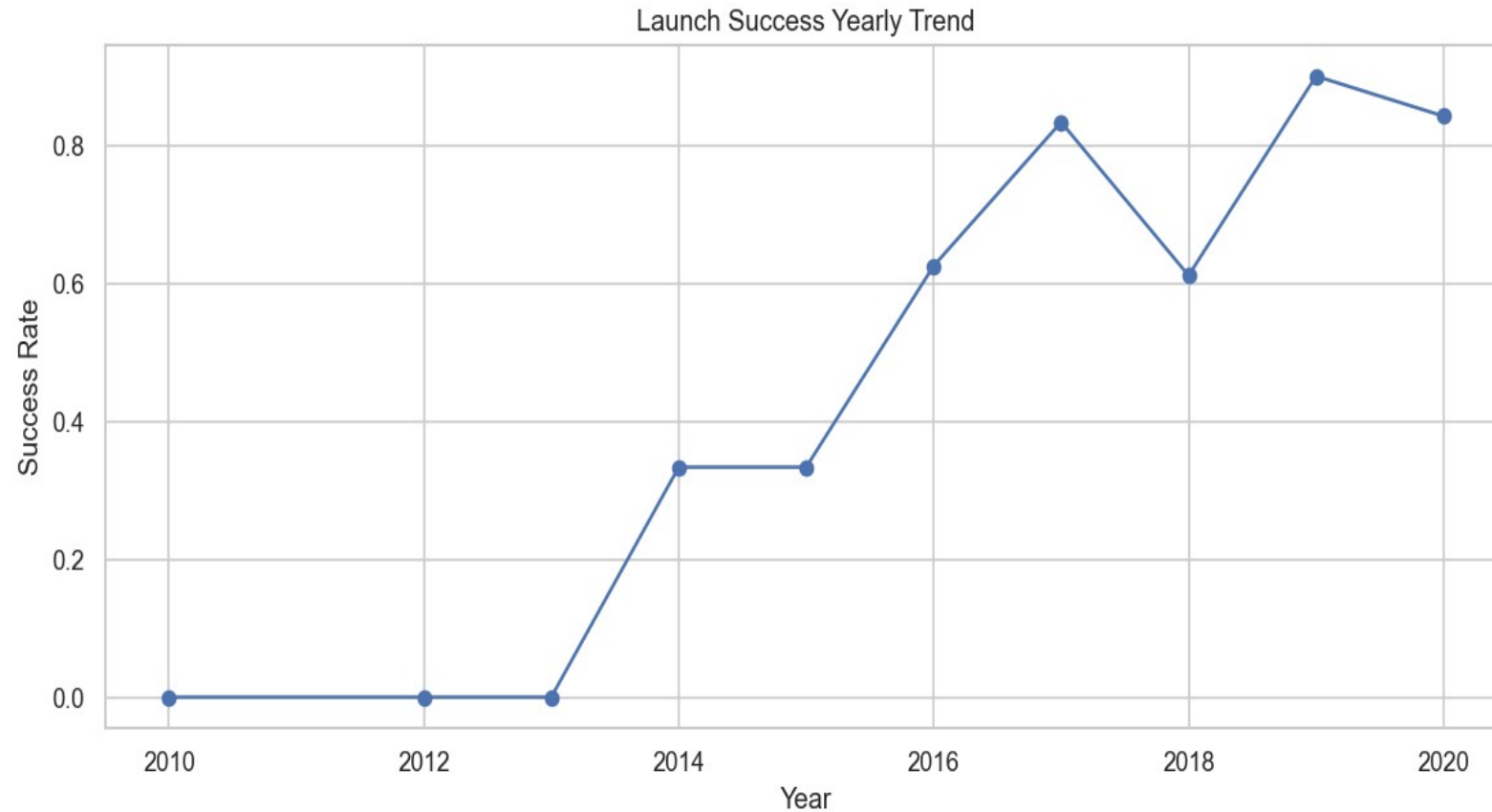
Orbits such as ES-L1, GEO, HEO and SSO show the highest landing success; GTO is among the weakest.

EDA Results — Flight Number vs Orbit



In LEO, success is linked to flight number; in GTO there is no clear relationship.

EDA Results — Yearly Success Trend



Success rate climbs steadily from 2013 onward as SpaceX accumulates landing experience.

EDA with SQL — Key Findings

4 launch sites

CCAFS LC-40, KSC LC-39A, VAFB SLC-4E, CCAFS SLC-40

619,968 kg

total payload carried by NASA (CRS) boosters

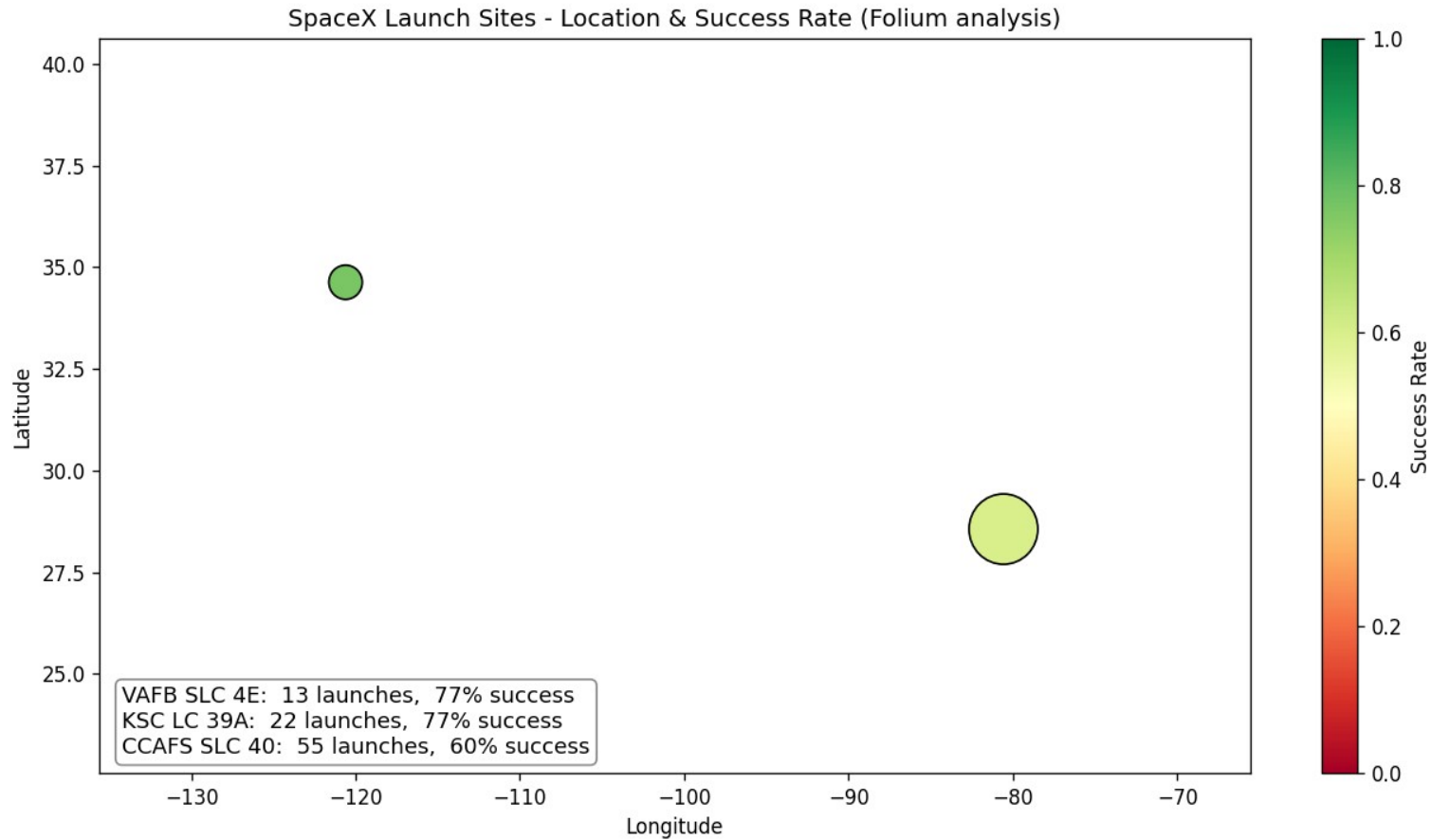
2015-12-22

first successful ground-pad (RTLS) landing

F9 B5 B1048+

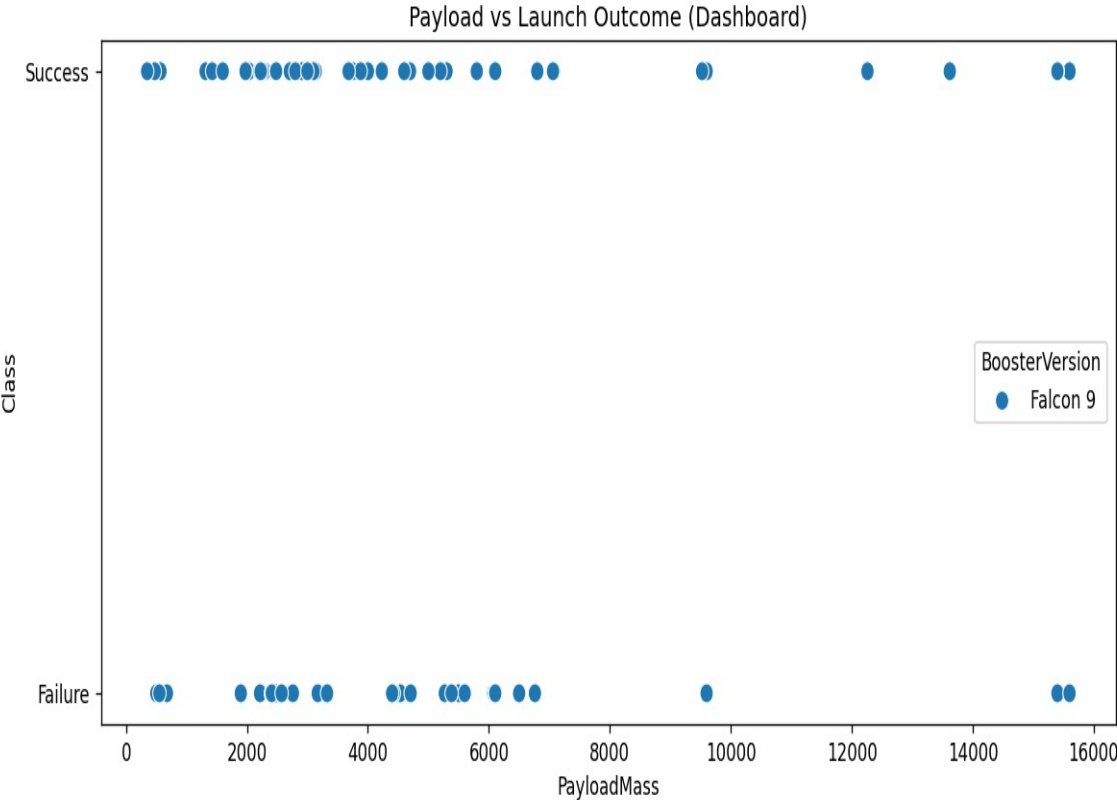
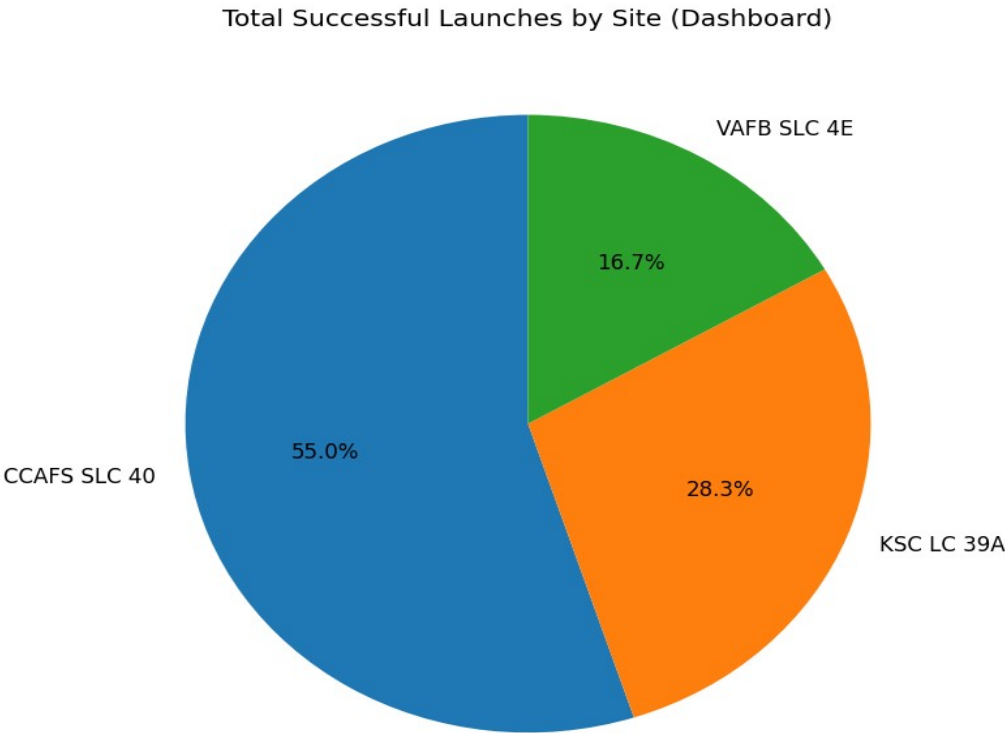
boosters carrying the maximum payload (15,600 kg)

Interactive Map — Folium



- All launch sites sit near the coast and the equator.
- Marker color = success rate, size = number of launches.
- Sites are close to railways and highways but away from cities.

Plotly Dash — Dashboard Results

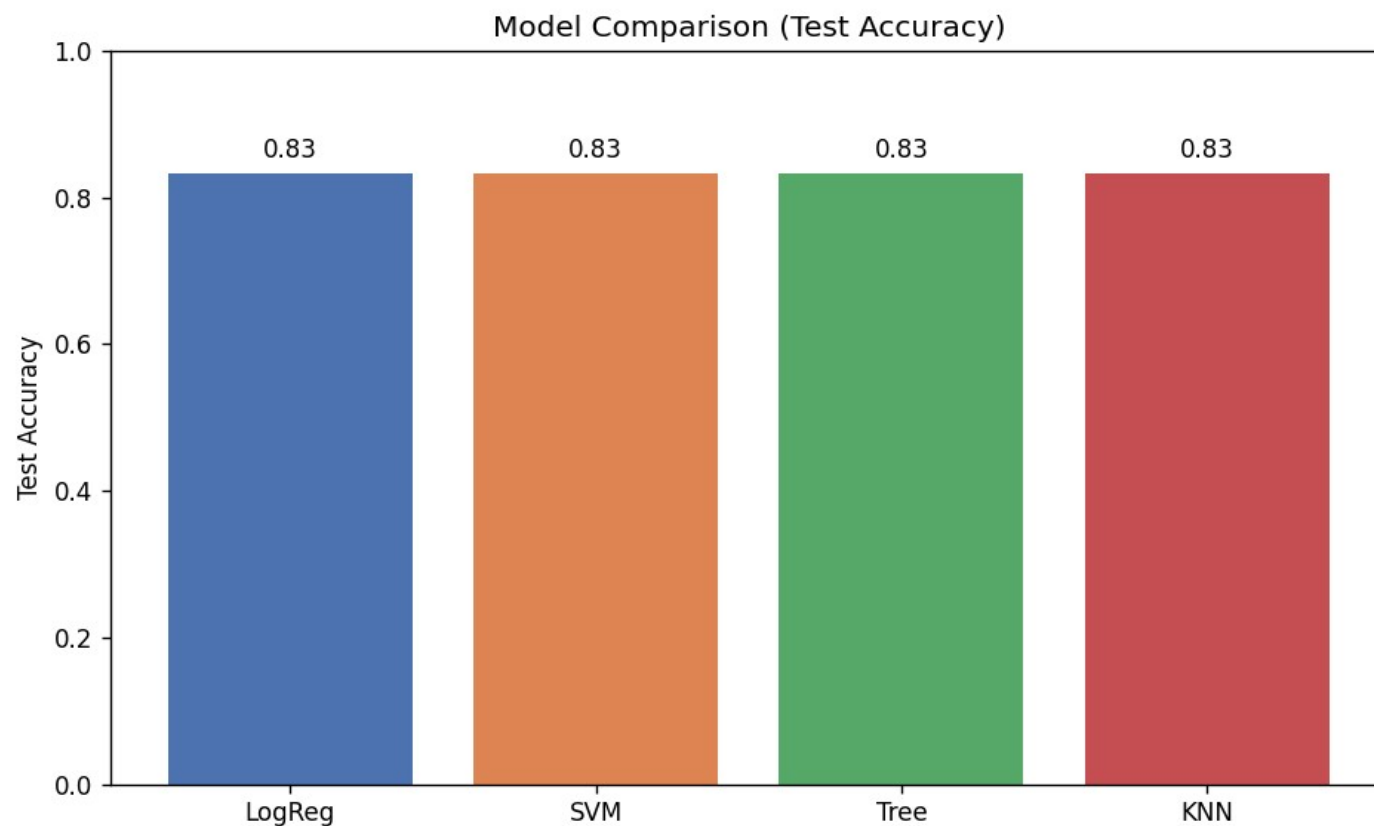


Success share by site (left) and payload vs outcome by booster version (right).

Predictive Analysis — Methodology

- Standardized the feature matrix and split into train / test (80 / 20, 18 test rows).
- Trained Logistic Regression, SVM, Decision Tree and KNN.
- Tuned hyperparameters with GridSearchCV (10-fold cross-validation).
- Selected the best model by accuracy and inspected the confusion matrix.

Predictive Analysis — Model Comparison

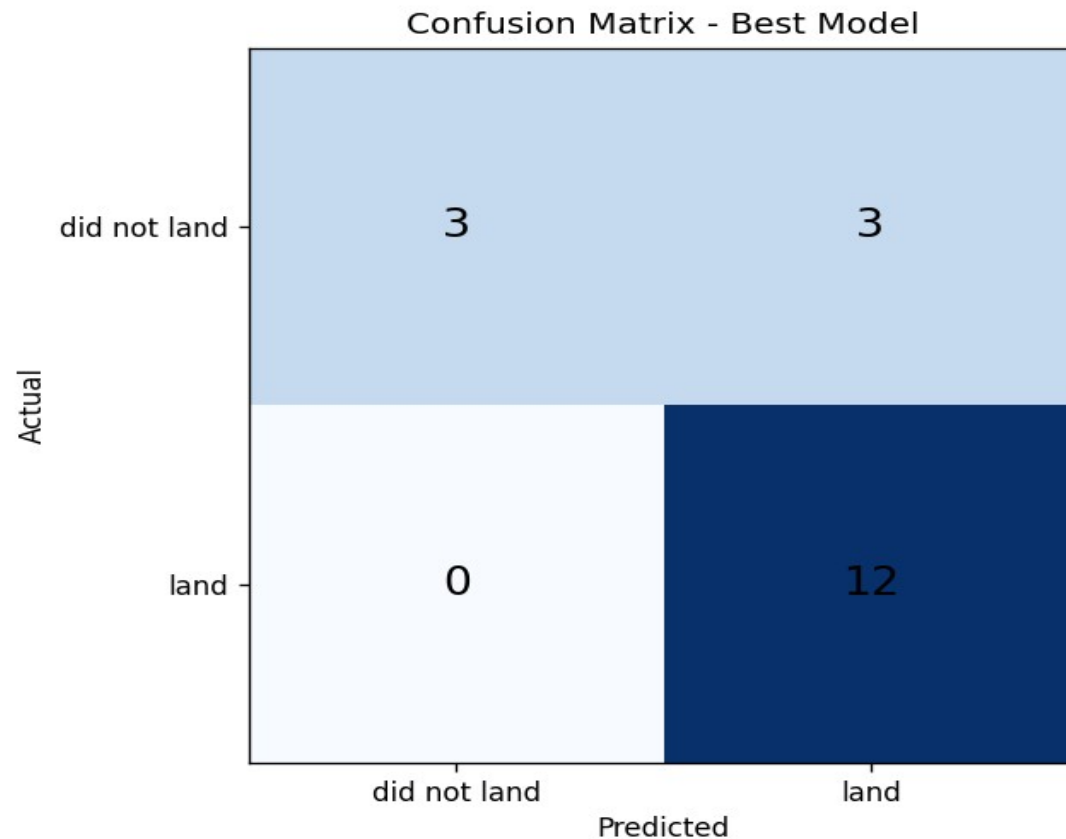


83.3%

test accuracy

All four models reach the same accuracy on the small test set; Decision Tree had the best validation score.

Predictive Analysis — Confusion Matrix



- The model correctly identifies most successful landings.
- The main weakness is false positives — predicting a landing that did not occur.
- More data would help separate the classes further.

Conclusions

- Landing success has risen markedly since 2013 as SpaceX gained experience.
- Flight number, payload mass, orbit type and launch site are all predictive of success.
- Low-weight orbits and certain sites (KSC LC 39A) show the highest success rates.
- All four tuned classifiers predict landings with ~83% test accuracy.
- Future work: enrich with weather and booster-reuse history, and gather more launches to reduce false positives.

Innovation: a deployable bid-estimation tool that turns landing probability into a launch-cost forecast.

Appendix & Resources

All notebooks (data collection, wrangling, EDA SQL, EDA viz, Folium, Dash, ML):

<https://github.com/Pointsd/Applied-Data-Science-Capstone>

Dataset: SpaceX REST API + Wikipedia launch records (90 Falcon 9 launches).

Thank you!